

Information Choice in Auctions[†]

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The choice of an auction mechanism influences which object characteristics bidders learn about and whether the object is allocated efficiently. Some object characteristics are valued equally by all bidders and thus are inconsequential for the efficient allocation. Others matter only to certain bidders and thus determine the bidder with the highest valuation. I show when the second-price auction is ex ante efficient by inducing bidders to seek socially relevant information. When facing a continuous learning trade-off, bidders learn more about socially relevant components and less about common characteristics of the object in a second-price auction than a first-price auction. (JEL D44, D83)

Preparing how to bid in an auction usually involves evaluating multiple characteristics of the object. This paper explores which object characteristics bidders gather information about before bidding. These issues are relevant to, for example, corporate takeovers, in which acquiring firms have access to a variety of information about a target company. This information includes the company's R&D activities and its stand-alone value. A reasonable assumption is that firms cannot perfectly process or uncover all existing information and are thus driven to select elements to focus on before the bidding takes place. Should an acquiring firm conduct research on characteristics that are specific to them, such as their R&D synergies with the target? Or should they focus on factors that also matter to other acquiring firms, such as the stand-alone value? Another example is resource rights auctions for oil fields or timber. Each bidder has access to the same uncertain volume of oil or timber when winning the auction. Bidders may incur different costs in extracting the resources from a site because of the use of different drilling or logging technologies. Does a bidder prefer to perform exploratory drilling to learn about oil volume, or to learn about the costs of extracting the resource through estimating the drilling costs specific to her, or some combination?

Allocative efficiency is of central importance in auction theory. Some characteristics are relevant only to individual bidders (henceforth, referred to as a *private component*), and they determine the efficient allocation. Others are relevant to all

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bidders (henceforth, a *common component*) and irrelevant for the efficient allocation. Different auction mechanisms might give rise to different incentives to bidders regarding which component to acquire information about.

As a simple example, consider two bidders who compete for one object. Each bidder's valuation is the sum of a common component S and her private component T_i , all drawn independently and uniformly on $[0, 1]$. Assume that (i) bidders can perfectly learn either their own private or the common component, but not both, and (ii) the bidder with the highest expected value wins the auction. If there exists an auction that incentivizes the bidders to learn about their private component, then the object is allocated efficiently with probability one. If an auction incentivizes bidders to learn only about the common component, the object is allocated to the lower-value bidder with probability one-half, achieving the same allocative efficiency as a random lottery.

The novelty of this paper lies in its illumination of *which* component of the object bidders seek to learn. This is in contrast to much of the auction theory literature, where the focus is on *how much* costly information bidders seek to acquire about one given component. In the first part of the analysis, bidders choose between costless experiments that, roughly speaking, generate the same amount of information about total object value but vary in their informativeness about the private component and the common one. At the extremes, an independent private values (IPV) framework arises if bidders learn exclusively about their private components, and a pure common value (CV) framework arises if bidders learn exclusively about the common component.¹ I show that the second-price auction (SPA) leads to ex ante efficient information acquisition and an IPV environment. The first-price auction (FPA), on the other hand, incentivizes bidders to learn about the common component at the expense of socially valuable private-component information. For a parametric example, I show when this rules out all but a pure CV framework in the FPA.

The second part of the paper analyzes a general continuous information choice model with costly experiments about both components. I extend the statistical concept of *accuracy* from Lehmann (1988) to rank real-valued signals in terms of their accuracy about several components.² I show that if higher accuracy about one component comes at the expense of lower accuracy about another, then the SPA leads to more information acquisition about the socially relevant private component and less information acquisition about the common component in comparison to a first-price auction. This extends a previous one-dimensional learning result by Persico (2000) to a multidimensional learning model and establishes the technical tools to compare experiments with varying accuracy about more than one random variable.

Bidders compete for a single indivisible object. Each bidder's valuation depends on two components: a common component (e.g., the stand-alone value of a firm) and an independent private value component (e.g., match-specific R&D synergies). Information choice—how much to learn about which component—is simultaneous and covert. Information plays a dual role: beyond containing information about the object, it is also informative about the signal of the opponent and hence her bid.

¹The IPV and CV settings lead to different theoretical predictions and vary significantly in their implications for auction design and policy. The literature on auctions typically assumes either one or the other at the outset of the analysis.

²As Lehmann (1988) showed, accuracy is a more complete order than the Blackwell order, which allows me to rank more experiments.

A rational bidder conditions her estimate of the object value not only on her own information but also on what she learns from the event of winning. In my model, the extent of the winner's curse and the interdependence between the bids are both endogenous and depend on the information choice. The signals of bidders become more interdependent if they learn more about the common component, and the winner's curse is exacerbated. If bidders learn only about their private component, then their information is independent of the other bidder—there is no winner's curse.

As a first baseline model, two bidders choose their signal from a set of costless signals, which vary in their informativeness about the two components, and each available signal leads to the same distribution of the bidder's expected object value. So, in a posted price mechanism, a bidder is indifferent between learning any available experiment. This allows me to focus on the strategic effect of learning about the two components and not the intertwined effect when bidders favor the more informative signal.

The first main result is that in an SPA, there exists a unique symmetric equilibrium in which bidders learn only about their private components. No resources are wasted by learning about the common component, which is irrelevant for efficiency, and the object is allocated to the bidder with the highest estimate of her private component. The intuition is that for any candidate equilibrium in which bidders learn to some extent about the common component, there exists a simple profitable deviation that decreases interdependence (by learning only about the private component). This deviation uses the fact that all signals have the same marginal distribution of the object's expected value: instead of learning anything about the common component, a bidder deviates to learning about her private component but continues to use the candidate-equilibrium bidding function. This deviation strategy does not change the marginal distribution of bids but makes the bidders' bid distributions independent.

A bidder's overall winning probability is not affected by this deviation; the probability that one of two identically and independently distributed random variables (bids) is higher than the other is one-half—the same as in any symmetric candidate equilibrium. How does the expected payment change with this deviation strategy? Conditional on winning in the SPA with the same marginal bidding functions, a bidder pays the second-order statistic of the two bids. How does the distribution of the second-order statistic vary with more or less correlation? At the extreme, under perfect correlation, the winning and losing bids coincide, and payment is the same as the winning bid. It is intuitive that as correlation decreases, the losing bid becomes more detached from the winning bid, and thus, expected payment decreases. I formalize this intuition as a general statistical property: the distribution of the second-order statistic of two correlated random variables (the distribution of the losing bid in the candidate equilibrium) dominates the distribution of the second-order statistic of two independent random variables (the distribution of the losing bid with the deviation) in terms of first-order stochastic dominance.³ By decreasing interdependence in the deviation, the distribution of the second-order statistic puts more weight on lower bids, and expected payment decreases. Finally, I show that the expected value

³This holds as long as these random variables have the same marginal distributions. The opposite is true for the distribution of the first-order statistic.

of the object conditional on winning is also weakly higher in the deviation than in the candidate equilibrium, establishing that it is indeed strictly profitable.

The second part of the paper analyzes a general model of information acquisition about two components, allowing for any number of bidders. Bidders choose how informative their one-dimensional private signal is about both the common component and their private one. Increasing the accuracy of the private signal about either component comes at higher monetary costs and might also influence the accuracy of the signal about the other component.

My second main result ranks learning incentives about either component between the FPA and the SPA in this general setup. I show that bidders in the SPA have lower incentives to learn common-component information as opposed to the FPA. If in addition to higher monetary costs, increasing accuracy about the private component comes at the opportunity cost of lower accuracy about the common component, then the SPA can be more efficient than the FPA since it also leads to higher incentives to learn about the private component than the FPA. This extends an insight by Persico (2000), who showed that in a one-component learning model, bidders learn more accurately in the FPA than in the SPA. This force also manifests in my model in that common-component information is more valuable in the FPA than the SPA. In addition, a new second force is present in my two-dimensional learning model because learning more about the common component has an impact on accuracy about the private component, and vice versa. In general, this affects the SPA and the FPA differently, convoluting the total effect of learning about either component. However, I show that in a symmetric candidate equilibrium, this effect of varying private-component information coincides in the SPA and the FPA. Hence, the difference in learning incentives about the two components in the SPA and the FPA is driven by the common-component accuracy.

Finally, I show that if learning about the private component goes hand-in-hand with learning about the common component, then learning about the private component is also more valuable in the FPA than in the SPA. Then, perhaps strikingly, bidders have higher incentives to learn about both components in the FPA than in the SPA. Consequently, the efficiency ranking between the FPA and the SPA reverses: for a numerical example, I show that the FPA achieves the highest feasible allocative efficiency, while the SPA outcome is as inefficient as a random lottery.

Related Literature.—In the classic literature in auction theory, the distribution of bidders' private information is exogenous and does not depend on the auction format. The literature on covert information acquisition in auctions⁴ endogenizes the private information of bidders by asking *how much* costly information they seek to acquire about a one-dimensional, payoff-relevant variable. In an IPV framework, Hausch and Li (1991) and Stegeman (1996) show that bidders' incentives to acquire information coincide in an FPA and an SPA.⁵ For a pure CV framework, Matthews

⁴Information acquisition has been analyzed in other areas of economics. See Bergemann and Välimäki (2002); Crémer, Spiegel, and Zheng (2009); Shi (2012); and Bikhchandani and Obara (2017) in mechanism design; Martinelli (2006) and Gerardi and Yariv (2007) in voting; Crémer and Khalil (1992) and Szalay (2009) in principal-agent-settings; and Rösler and Szentes (2017) in bilateral trade.

⁵See also Kim and Koh (2022) for flexible information acquisition in an IPV setup. For literature on *overt* information acquisition, see, e.g., Hausch and Li (1991) and Compte and Jehiel (2007). Hernando-Veciana (2009)

(1977) finds a condition on private signals for which bidders acquire the same amount of information about their common component in the SPA and the FPA.

The closest paper to mine is Persico (2000). In an affiliated-value model, bidders choose the accuracy (a ranking of informativeness by Lehmann 1988) of their signal about a one-dimensional random variable. Persico (2000) shows that bidders acquire more information in the FPA than in the SPA. The second part of my work in Section IIIB generalizes the techniques and findings in Persico (2000) to a two-dimensional learning model about two components.

Bergemann and Välimäki (2002) show that in a private value environment, the Vickrey-Clarke-Groves (VCG) mechanism leads to ex ante efficient information acquisition. This result does not cover my framework because values in my framework are interdependent as soon as any bidder learns about the common component. For an interdependent values setup, Bergemann and Välimäki (2002) provide conditions when agents acquire too much or too little information in comparison to an efficient benchmark, and they discuss the conflict between ex ante and ex post efficiency.⁶

My work also relates to the literature on mechanism design when agents face a costly private investment decision that affects their valuation instead of a costly learning decision about a fixed value.⁷ For a holdup problem, Rogerson (1992) shows that a VCG mechanism leads to an ex ante efficient level of investments. Crucially, in contrast to my framework, these results rely on an IPV environment in which an agent's investment has no externalities for other agents. See also Hatfield, Kojima, and Kominers (2018), who connect efficient and strategy-proof mechanisms with efficient incentives to invest. Neither auction format considered in my paper is strategy-proof due to the interdependent value environment.

My work also connects to the literature on auctions and mechanism design with private values when bidders have information not only about their own value but also the value of the other bidders, which is not directly payoff relevant for them (see, e.g., Fang and Morris 2006). Within this context, Gleyze and Pernoud (2023) allow agents to acquire costly information about their own and the others' values. They find that generically, an IPV framework does not arise, and bidders learn about others' values unless the mechanism satisfies a separability condition (which the SPA does not).⁸ Kim and Koh (2020) analyze the incentives of two bidders with binary signals to learn the rival's private information in an interdependent values framework.

compares the incentives to overtly acquire information in the English auction and the SPA, when bidders can learn about either a common component or a private component (which is exogeneously given).

⁶ See also Bergemann, Shi, and Välimäki (2009), where the value is a weighted sum of every bidder's type. Learning cannot introduce any dependence among the types of bidders since all types are independent.

⁷ See also Arozamena and Cantillon (2004) for investment incentives of one firm to reduce its costs in a procurement auction. All costs are drawn independently and investment is publicly, relating to the literature on overt information acquisition in an IPV framework. They show that in contrast to an SPA, the FPA induces inefficiently low investment incentives.

⁸ It is not the strategic interaction in the auction (i.e., anticipating the others' bids) but the optimal spending on learning costs that drives the difference between Gleyze and Pernoud (2023) and my model. Consider an SPA with private values, as in Gleyze and Pernoud (2023), and let all but one bidder learn only about their own values. For the remaining bidder, learning about others has no impact on the best-response bid but might save learning costs: if other bidders' valuations are so high that the bidder never wins, she would not want to engage in costly learning about her own value.

This paper is part of a research area on information acquisition about common or idiosyncratic aspects in other games and decision problems. Bobkova and Mass (2022) ask how agents in a social learning framework split their learning budget between a common-component experiment and an idiosyncratic-component experiment. Perego and Yuksel (2022) analyze the incentives of media outlets to disclose information about issues of common interest versus issues for which readers' preferences are heterogeneous. In a sender-receiver communication framework, Deimen and Szalay (2019) ask whether a sender learns about her own or the receiver's optimal action.

The interplay between learning about the common and private component in my model also bears resemblance to Morris and Shin (2002) and the literature on the social value of public versus private information in games with strategic complementarities. In Morris and Shin (2002), agents learn both a private signal and public one and use this information to predict an unknown state and actions of others. Due to the game's coordination motive, agents react too strongly to the public signal. In my model, the signal about the common component plays a related role: it facilitates coordination between bidders even though common-component signals are not perfectly correlated as is the public signal in Morris and Shin (2002). Moreover, a social planner prefers bidders to not learn the common-component signal if it comes at the expense of the socially relevant private component. My model substantially differs from Morris and Shin (2002). Most notably, in my model bidding functions do not exhibit strategic complementarities (see, e.g., Athey 2002), which leads to a fundamentally different strategic problem depending on the auction format. Moreover, in Morris and Shin (2002), all information is about the same one-dimensional state of the world. In my model, however, bidders choose about which component of the multidimensional state to learn. These parallels and differences with my work are also shared by the literature on information acquisition in games with strategic complementarities, e.g., Hellwig and Veldkamp (2009) and Myatt and Wallace (2012).

I. Model

There are $N \geq 2$ risk-neutral bidders competing for one indivisible object. The reservation value of the auctioneer and the outside options of the bidders are zero. The valuation for the object of bidder i , denoted by V_i , is a function of a common component S and her private component T_i , $V_i = \psi(S, T_i)$, where $\psi(\cdot, \cdot)$ is a continuously differentiable, nonnegative, and strictly increasing function. The common component S is identical for every bidder and admits a density $h_S(\cdot)$ on its full support $\mathcal{S} \subset \mathbb{R}^+$; the private component T_i admits a density $h_T(\cdot)$ on its full support $\mathcal{T} \subset \mathbb{R}^+$. The common component and all private ones $\{S, T_1, T_2, \dots, T_N\}$ are mutually independent, and $T_1, \dots, T_N$ are drawn identically.

The bidders do not observe the realizations of any component. Instead, they can learn about the common component and their private one by choosing one experiment in a set of real-valued experiments $X_i \in \mathcal{E}$.⁹ Bidders choose their experiment

⁹In what follows, I impose further restrictions on the set of experiments and their costs. Up to Section IIIA, bidders choose from a set of free experiments, while in Section IIIB bidders choose the (costly) accuracy of their signal about both components.

X_i knowing the auction format. Information choice is covert: bidders do not observe which experiment their opponent has learned. Signals are independent conditional on the common component S . The game consists of two stages. After an auction format (SPA or FPA) is announced, bidders simultaneously choose their experiment in the *information-choice stage*. Thereafter, in the *auction stage*, bidders privately observe the realization of their experiment and bid in the auction.

I analyze symmetric monotonic Bayesian Nash equilibria in pure strategies (for brevity referred to as equilibria in what follows) for the FPA and the SPA, in which bidders choose the same experiment X^A and use an increasing bidding function $\beta_A(x)$ for auction format $A \in \{I, II\}$.¹⁰

II. Motivating Example

To set intuition, it is instructive to consider a stylized model with $N = 2$ bidders and a simple parametric information choice absent any monetary learning costs. The three components, T_1, T_2, S , are drawn independently and equiprobably from $\{0, 1\}$. The valuation of bidder $i \in \{1, 2\}$ for the object is $V_i = S + T_i$. Bidders choose from a set of experiments parameterized by $\sigma \in [\underline{\sigma}, \bar{\sigma}] \subseteq [0, 1]$, where experiment X_i^σ with support on $[0, 1]$ has the following densities:

$$f^\sigma(x|S = 0, T_i = 0) = 2 - 2x;$$

$$f^\sigma(x|S = 1, T_i = 0) = \sigma 2x + (1 - \sigma)(2 - 2x);$$

$$f^\sigma(x|S = 1, T_i = 1) = 2x;$$

$$f^\sigma(x|S = 0, T_i = 1) = \sigma(2 - 2x) + (1 - \sigma)2x.$$

For any choice of σ , the experiment is more likely to produce high signal realization when both components are low and low signal realizations when both components are high. However, when one payoff-relevant component is high and the other is low, the choice of σ determines which signal realizations are more likely: for example, with a high common component and a low private one, $\sigma > 1/2$ makes higher signals increasingly likely, while $\sigma < 1/2$ makes lower signals increasingly likely.

This signal technology introduces a learning trade-off between the common component and the private one, governed by the choice of σ : if feasible, $\sigma = 1$ yields the most informative signal exclusively about the common component but no information about the private component (the marginal signal distribution is uniform on $[0, 1]$ for all $T_i \in \{0, 1\}$); if feasible, $\sigma = 0$ is the most informative signal about T_i , while containing no information about the common component. The higher σ , the more informative the signal about S and the less informative the signal about T_i .¹¹ The case $\sigma_1 = \sigma_2 = \underline{\sigma}$ plays a special role in the following analysis: it is the

¹⁰ See the online Appendix for a discussion of asymmetric equilibria and mixed strategies.

¹¹ See Section IIIB for a definition of informativeness used in this paper: *accuracy*, following Lehmann (1988) and Persico (2000). Formally, for this example, for any $\sigma' > \sigma$, $X_i^{\sigma'}$ is (i) more accurate about S and (ii) less accurate about T_i than X_i^σ but (iii) equally accurate about V_i .

learning outcome in any ex ante efficient equilibrium.¹² When $\sigma \in [0, 1]$, then the model nests two well-known frameworks. If $\sigma_1 = \sigma_2 = 1$, then the environment is a *pure CV framework*. If $\sigma_1 = \sigma_2 = 0$, then this is an *IPV framework*.

In contrast to learning about individual components, bidders do not face a trade-off in how informative their signal is about V_i : for any choice of σ , the unconditional signal distribution of X_i^σ is uniform on $[0, 1]$, and the expected value $E[V_i | X_i^\sigma = x] = 1/2 + x$ is independent of σ . Hence, in a nonstrategic environment with a posted price for the object, a bidder would be indifferent between any σ . However, in the strategic auction environment, choosing a higher σ might yield a more correlated signal with the other bidder at the expense of lower information rents.

Let (σ, β_A^σ) for $A \in \{I, II\}$ be a candidate equilibrium in the FPA or SPA, where bidders choose σ and use an increasing bidding function β_A^σ .¹³ Without loss, I discuss bidder 1's profitable deviation in this candidate equilibrium. Bidder 1's expected ex ante utility when her information choice is σ_1 , she bids optimally after learning $X_1^{\sigma_1}$ and when bidder 2 plays the candidate equilibrium (σ, β_A^σ) , is denoted by $R^A(\sigma_1, \sigma)$. With this notation, the main result of this section follows: learning less (more) about the common component and more (less) about the private one is strictly profitable in the SPA (FPA), which substantially reduces the set of possible equilibria.

PROPOSITION 1: *For any $\sigma_1 < \sigma$, $R^{II}(\sigma_1, \sigma) > R^{II}(\sigma, \sigma)$. For any $\sigma_1 > \sigma > 0$, $R^I(\sigma_1, \sigma) > R^I(\sigma, \sigma)$.*

In any symmetric pure strategy equilibrium of the SPA, $\sigma = \underline{\sigma}$. In any symmetric pure strategy equilibrium of the FPA, $\sigma = \bar{\sigma}$ if $\underline{\sigma} > 0$, and $\sigma \in \{0, \bar{\sigma}\}$ otherwise.

When bidder 1 deviates to $\sigma_1 \neq \sigma$, then what she learns and how she bids affects both her expected payment as well as her winner's curse. For the stylized example of this section, there is a simple class of deviation strategies that dramatically simplifies the calculus of profitable deviations. It exploits the effect of deviations on the expected payment, while not affecting the degree of the winner's curse: *use the same bidding function $\beta_A^\sigma(\cdot)$ after deviating to $\sigma_1 \neq \sigma$* . Then, after deviating, bidder 1 still wins if and only if she has the higher signal realization.

With such a deviation strategy, at the lowest value $V_1 = 0$ (i.e., when $S = T_1 = 0$) or at the highest possible value $V_1 = 2$ (i.e., when $S = T_1 = 1$), bidder 1's information choice σ_1 is inconsequential since it does not affect her signal's density. However, when the value is $V_1 = 1$, then increasing σ_1 shifts bidder 1's probability of having the highest signal—and hence, her winning probability—from the event $(S = 0, T_1 = 1)$ to the event $(S = 1, T_1 = 0)$, without changing her overall winning probability. Because bidder 1 wins at every value with the same probability,

¹² Both bidders share the same common component, and the values are nonnegative. Thus, allocative efficiency requires the object to go to the bidder with the highest expected private component. Ex ante efficiency in this environment requires that bidders learn as much as feasible about their private components and the bidder with the highest signal, i.e., the highest expected private component, wins.

¹³ For any fixed information choice $\sigma = \sigma_1 = \sigma_2$, see, e.g., Milgrom and Weber (1982) for the equilibrium bidding functions in the FPA and the SPA. I omit them here since to establish Proposition 1, it is sufficient to assume only that the bidding functions are increasing.

such deviations have no effect on bidder 1's winner's curse. There is, however, an effect on the expected payment in the FPA and the SPA. The higher σ_1 , the more correlated are the signals of bidders 1 and 2, and hence, the higher is the expected payment of a bidder in the SPA, and the lower is the expected payment in the FPA. This is because intuitively, in the FPA, a higher correlation with the opponent helps a bidder to "leave not too much money on the table" by bidding closer to the losing bid, and lower correlation in the SPA helps a bidder to decouple her winning bid from the losing bid that she has to pay.¹⁴

The main insight is that when available signals do not vary in their informativeness about the valuation overall but only in their component-wise information content, then the incentives in the SPA push bidders toward the lowest feasible degree of correlation (if available, the IPV outcome), and the incentives in an FPA push them toward the highest feasible degree of correlation in private signals (if available, the pure CV outcome). Higher correlation means learning more about the common component at the expense of private-component information and hence to the detriment of allocative efficiency. So, the SPA is more efficient than the FPA.

When $\sigma = 0$, bidder 1 cannot affect her signal's interdependence with that of her opponent; signals are independent, have the same marginal distribution, and contain the same amount of information about V_i . Therefore, an equilibrium with $\sigma = 0$ exists by construction in both the SPA and the FPA. Note that in the FPA, the equilibrium with $\sigma = 0$ is not robust in the following sense: if introducing a slight degree of correlation between bidders' signals such that bidder 1 can influence her signal's correlation with that of bidder 2 (e.g., if the common and private components are slightly correlated or if $\underline{\sigma} > 0$ and $\underline{\sigma} \rightarrow 0$), then deviating to more correlation from $\underline{\sigma}$ is strictly profitable, leaving only the highest-correlation outcome $\bar{\sigma}$ in any equilibrium.

III. Generalization

This section shows that the previous insights extend beyond the parametric example of Section II. In Section IIIA, I show that the SPA is the ex ante efficient auction format when bidders choose between costless experiments that generate the same amount of information about the total value. In Section IIIB, I show that the SPA leads to more information acquisition about the private component and less about the common one in a general class of environments (but not all).

A. IPV in the SPA

Consider $N = 2$ bidders with additive valuation $V_i = S + T_i$. Each bidder chooses one experiment from the set $\mathcal{E}$ absent monetary costs. This is a general set of experiments, allowing bidders to learn possibly about both components

¹⁴Figure 1 depicts this effect for $\sigma = 1$. The blue lines show the distribution of bidder 2's signal (whose bid bidder 1 pays in an SPA) conditional on bidder 1 having a higher signal, when $\sigma_1 = 1$ for the solid line, and in a deviation with $\sigma_1 = 0$ for the dashed line. Similarly, the green lines show the distribution of bidder 1's signal (whose bid bidder 1 pays in an FPA) conditional on bidder 1 having a higher signal, when $\sigma_1 = \sigma = 1$ for the solid line, and when deviating to $\sigma_1 = 0$ for the dashed line.

simultaneously without imposing parametric restrictions or order on the experiments. The following two assumptions impose more structure on the experiments.

ASSUMPTION 1: *The set of experiments $\mathcal{E}$ satisfies*

- (i) $X_1 \perp\!\!\!\perp X_2 | S$, $X_1 \perp\!\!\!\perp T_2$, and $X_2 \perp\!\!\!\perp T_1$ for any X_1 and X_2 ;
- (ii) for each i , there exists a unique private-component experiment $X_i^T \in \mathcal{E}$ that is drawn independently of S and satisfies the strong monotone likelihood ratio property (MLRP) with respect to T_i ;
- (iii) for all $X_i \in \mathcal{E} \setminus \{X_i^T\}$, X_i satisfies the strong MLRP with respect to S and the weak MLRP with respect to T_i .

Assumption 1 controls the correlation between the bidders' experiments. By (i), any two experiments are independent conditional on the common component S , and independent from the other's private component. By (ii), there exists an experiment that is informative only about one's private component and independent from the other bidder's signal. By (iii), for any experiment that is not purely a private-component experiment, higher signal realizations are more indicative of higher realizations of the common component.

Assumption 2 defines "the same amount of information" to allow a comparison between experiments: they lead to the same distribution of the expected object value.

ASSUMPTION 2: *Any two experiments $X_i \in \mathcal{E}$ and $X_i' \in \mathcal{E}$ lead to the same distribution of the expected value of the object, denoted by $E[V_i | X_i] \stackrel{d}{=} E[V_i | X_i']$. In particular,¹⁵*

- (i) X_i and X_i' are distributed i.i.d., and
- (ii) for all $x \in [0, 1]$, $E[V_i | X_i = x] = E[V_i | X_i' = x]$.

This assumption is clearly satisfied for the parametric example in Section II. Generally, if all components are distributed identically, then any private-component experiment X_i^T with marginal distribution $F^T(x | t_i)$ and any common-component experiment X_i^S with marginal distribution $F^S(x | s)$ with $F^S(x | s) = F^T(x | t)$ for $s = t_i$ satisfy $E[V_i | X_i^S] \stackrel{d}{=} E[V_i | X_i^T]$. As the following example shows, there are several other nonsymmetric examples that satisfy Assumption 2.

EXAMPLE 1: Let $S \in \{0, 100\}$ with equal probability, $T_i \sim \mathcal{U}[0, 100]$, and $V_i = S + T_i$. The private-component signal is fully revealing, $X_i^T = T_i/100$. The common-component signal has densities $f^S(x | S = 0) = 2 - 2x$ and $f^S(x | S = 100) = 2x$. For example, for a signal realization $X_i^S = 0$, the posterior value is distributed uniformly on $[0, 100]$, while for a signal realization $X_i^T = 0$,

¹⁵ This is without loss after relabeling, as both signals X_i^S and X_i^T have the same full support.

the posterior value is 0 or 100 with equal probability. Importantly, in both cases, the expected value $E[V_i | X_i^S = 0] = E[V_i | X_i^T = 0] = 50$ coincides, while the distribution of the posterior value can differ between the two signals.

If Assumption 2 holds and a bidder is offered the object for a posted price, then she is indifferent between learning any experiment in the set $\mathcal{E}$: they all lead to the same distribution of the expected object value, and the component composition beyond its inference about V_i is payoff-irrelevant. However, in a strategic environment, learning certain experiments as opposed to others might give a bidder a better idea of her opponent's bid. Thus, it need not be the case that a bidder is indifferent between signals in $\mathcal{E}$ even when Assumption 2 is satisfied. This is due to the strategic effects of bidders choosing more or less correlated signals and not due to one signal containing more information about the object's value.

The following result shows that even though bidders can learn about both components simultaneously, they choose not to do so in an SPA. Thus, IPV arises endogenously even with a much richer nonparametric signal structure than in Section II.

PROPOSITION 2 (IPV in the SPA): *If Assumptions 1 and 2 hold, then in any symmetric pure strategy equilibrium in the SPA, bidders acquire information only about their private component by learning X_i^T .*

The bidder with the highest signal realization about her private component wins, and bidders do not waste their learning resources on the common component, which is irrelevant for the optimal allocation. The remainder of this section is devoted to providing intuition for why only an IPV framework arises endogenously in the SPA.

The existence of an IPV equilibrium follows from Assumption 2. If bidder 2 learns X_2^T , then there is no winner's curse for bidder 1. If it is optimal for bidder 1 to bid b after learning $X_1^T = x$, then bidding b is also optimal when learning $X_1' = x$ for any other $X_1' \in \mathcal{E}$. This is because all available signals with realization x lead to the same expectation of V_1 , and all are independent of bidder 2's bid. All signals in $\mathcal{E}$ are distributed identically, so bidder 1 is indifferent between learning either since they all lead to the same decision problem, given a bidding strategy of bidder 2.

Why is there no equilibrium in which both bidders learn anything about the common component? By contradiction, consider a candidate equilibrium where bidders learn $X_i' \neq X_i^T$ and use an increasing bidding function $\beta(\cdot)$. The following constitutes a strictly profitable deviation for bidder 1: *learn X_1^T instead of X_1' , and bid $\hat{\beta}(x) = \beta(x)$ for all x .* I show that this deviation leads to (i) the same winning probability, (ii) a strictly lower expected payment, and (iii) a weakly higher expected value when winning. Hence, a bidder is better off using the same marginal bid distribution as in the candidate equilibrium but applied to her private-component signal. Let $F'(x|s)$ be the distribution of experiment X_i' conditional on $S = s$, and $F(x)$ be the marginal distribution of both X_i' and X_i^T (and any other experiment in $\mathcal{E}$ by Assumption 2).

(i) *Same winning probability.* In the symmetric candidate equilibrium, both bidders win with probability $1/2$. If bidder 1 deviates, then the bidders' signals are independent. Recall that X_1' and X_1^T are distributed identically, and both lead to the

same bid distribution in the deviation strategy. Hence, both bidders have the same marginal bid distribution and win when they have the higher signal. Thus, the deviation has no effect on bidder 1's winning probability:

$$\Pr(X_1^T \geq X_2') = \int [1 - F(x)] dF(x) = \frac{1}{2}.$$

(ii) *Strictly lower expected payment.* Let $G_{(2)}(x) := \Pr(X_2' \leq x | X_1' \geq X_2')$ be the distribution of the signal of bidder 2 in the candidate equilibrium, conditional on bidder 1 winning. This is the distribution of the second-order statistic of the two correlated, identically distributed signals, which is as follows:

$$G_{(2)}(x) = 2F(x) - \int_S F'(x|s)^2 h_S(s) ds.$$

Let $\hat{G}_{(2)}(x) := \Pr(X_2' \leq x | X_1^T \geq X_2')$ be the distribution of bidder 2's signal conditional on bidder 1 winning when deviating. Recall that signals X_2' and X_1^T are i.i.d. Then, $\hat{G}_{(2)}(x)$ equals the well-known distribution of the second-order statistic of two i.i.d. random variables:

$$\hat{G}_{(2)}(x) = 2F(x) - F(x)^2.$$

When is bidder 1's payment lower? In both the candidate equilibrium and the deviation, bidder 1 pays the bid of bidder 2, who uses the bidding function β . If bidder 1 wins, she pays $\int \beta(x) dG_{(2)}(x)$ in the candidate equilibrium and $\int \beta(x) d\hat{G}_{(2)}(x)$ in the deviation. As I show in the Appendix, $G_{(2)}$ and $\hat{G}_{(2)}$ can be ranked:¹⁶ $G_{(2)}$ first-order stochastically dominates $\hat{G}_{(2)}$. In the deviation, bidder 2's signal realizations, and hence, her bids, are more likely to be distributed lower when bidder 1 wins. Since the bidding function β is increasing, it follows that the expected payment is also lower. Consider the extreme case of perfectly correlated signals in the candidate equilibrium. Then, both bidders bid the same, the first- and second-order statistics of the bids coincide, and bidder 1 pays her own bid in the SPA when winning. However, in the deviation, bidder 1 pays less conditional on winning because the two bids no longer coincide: there is a gap between the distribution of the second-order statistic of the bids (which bidder 1 pays when winning) and the first-order statistics of the bids (which bidder 1 bids when winning). The deviation increases the gap between the first- and the second-order statistics distributions. This increase takes the form of first-order stochastic dominance, as depicted in Figure 1 for the motivating example in Section II. Therefore, with any strictly increasing bidding function β , the expected payment for bidder 1 is strictly lower in the deviation than in the candidate equilibrium.

(iii) *Higher value when winning.* While bidder 1's deviation strategy does not change her overall winning probability, it affects her exposure to the winner's curse by changing her winning probabilities across different component combinations.

In the symmetric candidate equilibrium, bidder 1 wins at every S with probability $1/2$. Hence, the common-component payoff conditional on winning is

¹⁶ See Lemma 2 in Appendix A.A for a general ranking of the distributions of first- and second-order statistics of two variables with the same marginal distribution and varying correlation.

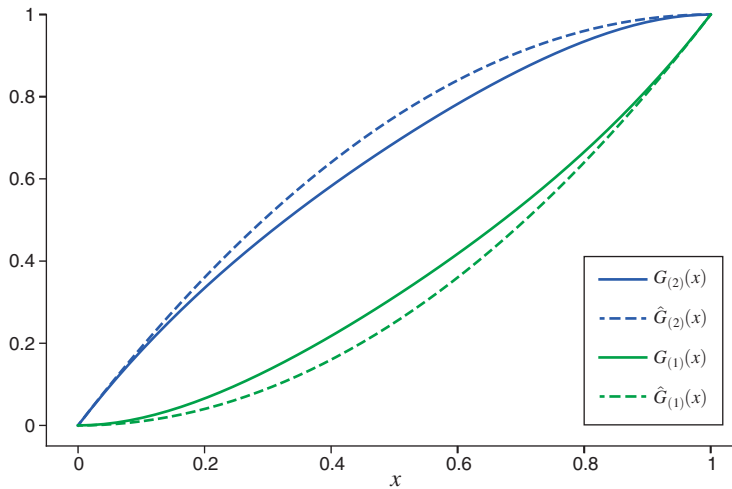


FIGURE 1. DISTRIBUTIONS OF FIRST- AND SECOND-ORDER STATISTICS OF TWO IDENTICALLY DISTRIBUTED RANDOM VARIABLES

$E[S|X'_1 \geq X'_2] = E[S]$. When deviating, bidder 1 no longer wins with probability $1/2$ at every realization of S . When bidder 1 deviates, bidder 2 (the only bidder now informed about S) is more likely to win when X'_2 is high. Hence, conditional on winning, bidder 2's expected value of S is higher than in the candidate equilibrium. Since the object is always sold (i.e., no surplus is destroyed), the two bidders are dividing among themselves the total surplus from the common component, $E[S]$. Both bidders are equally likely to win. Hence, bidder 2's expected common component when winning increases, and bidder 1's common-component value when winning decreases by the same amount.

In addition, bidder 1's expected private component is higher in the deviation than in the candidate equilibrium due to two effects. First, the higher realizations of X'_1 at which bidder 1 is more likely to win are even better news about the private component than the same signals in the candidate equilibrium (this is due to Assumptions 1 and 2). If bidder 1's distribution of winning signals (the first-order statistic of the two bidders) was the same in the candidate equilibrium and the deviation strategy, then this increase would exactly offset the loss of bidder 1 from winning at a lower common component due to Assumption 2 and symmetry in marginal bid distributions. However, the distribution of bidder 1's winning signal in the deviation (the first-order statistic of two independent signals) first-order stochastically dominates that in the candidate equilibrium (the first-order statistic of two correlated signals). This raises the expected private component when winning even further, outweighing the loss from the expected common component when winning.

Discussion and Extensions.—A corollary of Proposition 2 is that the SPA is *ex ante efficient*: among all pure strategy symmetric equilibria, the SPA leads to the highest expected object value for the winner. This is because each bidder is equally likely to win at any realization of the common component, and the expected private component of the winning bidder is highest when bidders learn only about their

private components as induced by the SPA, $E[T_1 | X_1^T \geq X_2^T] \geq E[T_1 | X_1 \geq X_2]$ for all experiments $X'_i \in \mathcal{E}$.¹⁷

In the online Appendix, for a simplified framework in which bidders learn either about the common component or their private one, I discuss how robust are the insights of Proposition 2 and under which conditions the SPA loses its ex ante efficiency properties. I provide sufficient conditions when Proposition 2 extends to an arbitrary number of bidders (Propositions 9 and 10). In Proposition 6, I show that even if bidders are allowed to play mixed learning strategies, they learn only about their private component in an SPA. Dropping Assumption 2, I show how to extend Proposition 2 beyond experiments with the same distribution of the expected value (Proposition 15), and when such an extension is not possible. Among other extensions, I also discuss asymmetric equilibria and two-dimensional signals in the online Appendix.

It is instructive to consider in an FPA the effect of the same deviation strategy for bidder 1 from a candidate equilibrium where bidders learn $X'_i \neq X_i^T$. By the same argument as for the SPA, bidder 1's winning probability in an FPA will be one-half. In an FPA, the winning bidder pays her own bid and hence the first-order statistic of the two bids. The distribution of the first-order statistic of the two signals (and hence bids) in the deviation first-order stochastically dominates that of the candidate equilibrium. Here, a higher correlation in the candidate equilibrium as opposed to the deviation strategy corresponds to leaving "less money on the table" by bidding closer to the opponent. This shows that in contrast to the SPA, the same deviation in the FPA leads to a strictly higher expected payment. However, similarly to the SPA, in the deviation strategy learning about one's private component can lead to more information rents and hence gains in terms of the expected value when winning. Assumption 2 is not sufficient to pin down which aspect of the deviation strategy dominates in the FPA—the higher expected payment or the higher expected value. By the same argument, even if a deviation to a pure common-component signal were available to bidder 1, Assumption 2 would not pin down whether the lower expected payment by anticipating the opponent's bid better would offset a loss in information rents of the private component. Hence, under Assumption 2, the result from the motivating example does not generalize to the same extent for the FPA as for the SPA.

B. Costly Information Choice

Can the insights of Section IIIA be interpreted as a microfoundation for an IPV assumption in the SPA? A crucial assumption in Proposition 2 is that it abstracts from any monetary costs of learning, and every available experiment leads to the same distribution of the expected value. This reduces information choice purely to strategic considerations and not to decisions about how much to learn about V_i . In reality, it might be that bidders can more easily learn a bit about both components than a lot about one component. A bidder might then prefer a signal that is simultaneously informative about both components.¹⁸ Moreover, an IPV framework cannot arise

¹⁷ See Lemma 3 in the Appendix for a formal result.

¹⁸ As an extreme illustration, consider a (possibly) fully revealing signal $X_i^p = \rho_i S + (1 - \rho_i)T_i$. While $\rho_i \in \{0, 1\}$ perfectly reveals one component, $\rho_i = 1/2$ fully resolves all uncertainty about V_i .

if a private-component signal X_i^T is not available or if learning costs satisfy some Inada conditions such that learning marginally about the common component has zero marginal cost. Hence, in general, an IPV framework is unlikely to arise endogenously if bidders consider both the strategic implications, the informativeness, and the costs of their experiments. For those cases, this section provides a relative result—a comparison of learning incentives in the SPA and the FPA.

To rank experiments in their informativeness in what follows, I use the notion of *accuracy* (Lehmann 1988; Persico 2000), which is commonly used in the auction literature. Accuracy is a more complete order than the Blackwell order. Moreover, when signals satisfy the MLRP and payoffs satisfy a single crossing property—both often the case in the auction literature—then a higher-accuracy signal results in a higher payoff for a decision-maker than a lower-accuracy signal.¹⁹

DEFINITION 1 (Accuracy): Fix a random variable Z and two real-valued signals X^a and X^b with corresponding continuous distributions $F_a(\cdot | z)$ and $F_b(\cdot | z)$. X^a is more accurate than X^b about Z (write $X^a \succ_Z X^b$) if for every x , the function

$$M(x|z) := F_a^{-1}(F_b(x|z)|z)$$

is nondecreasing in z . X^a and X^b are equally accurate about Z (write $X^a \sim_Z X^b$) if $X^a \succ_Z X^b$ and $X^b \succ_Z X^a$.

The above definition varies slightly from the standard definitions (e.g., those in Lehmann 1988) since it defines equal accuracy $\sim_Z$ and includes the variable Z explicitly in the binary relation $\succ_Z$. This captures that in my framework, Z can be several random variables. For example, two signals X^a and X^b can vary in their accuracy about the value V_i but also in their accuracy about the private component T_i .²⁰ The transformation $M(x|z)$ maps a signal realization x from the less accurate experiment to a signal realization for the more accurate experiment: the more accurate signal X^a is more likely to produce lower signals when Z is low, and high signals when Z is high, since it varies more strongly with Z than X^b .

Consider the general model in Section I for an arbitrary number of bidders with a valuation $V_i = \psi(S, T_i)$. Bidder i 's information choice is parameterized by two variables, $\sigma \in [0, 1]$ and $\tau \in [0, 1]$, so that set of experiments is $\mathcal{E} = \{X_i^{\sigma, \tau}\}_{(\sigma, \tau) \in [0, 1]^2}$. The cost of the information choice is $c(\sigma, \tau)$, where $c : [0, 1]^2 \rightarrow \mathbb{R}^+$ is a strictly increasing and continuously differentiable information cost function.

ASSUMPTION 3: $X_i^{\sigma', \tau} \succ_S X_i^{\sigma, \tau}$ for all $\sigma' \geq \sigma$, and $X_i^{\sigma, \tau'} \succ_{T_i} X_i^{\sigma, \tau}$ for all $\tau' \geq \tau$.

The information choice variable σ parameterizes the accuracy about the common component, and τ parameterizes the accuracy about the private component. A higher parameter (fixing the other parameter) corresponds to a more accurate signal

¹⁹ See the online Appendix for a formal result and an application of this to my two-dimensional learning environment for an extension of Proposition 2.

²⁰ In a decision problem with a single payoff-relevant variable, this additional notation is superfluous: two signals that are equally accurate about a single variable Z are essentially equivalent. Moreover, observe that if $X_i \sim_{V_i} X'_i$, then $E[V_i | X_i] \stackrel{d}{=} E[V_i | X'_i]$, but not vice versa.

about that component.²¹ For any information choice (σ, τ) , the signal depends only on the common component and the bidder's own private component. It has full support on $\mathcal{X}$, where $f^{\sigma, \tau}(x|s, t_i)$ and $F^{\sigma, \tau}(x|s, t_i)$ denote the signal's density and distribution, respectively.²² The marginal density of $X^{\sigma, \tau}$ for $S = s$ is denoted by $f_S^{\sigma, \tau}(x|s) := \int_T f^{\sigma, \tau}(x|s, t_i) dH_T(t_i)$, and $F_S^{\sigma, \tau}(x|s)$ is the corresponding distribution.

ASSUMPTION 4: *The set of experiments $\mathcal{E} = \{X^{\sigma, \tau}\}_{(\sigma, \tau) \in [0, 1]^2}$ satisfies the following:*

- (i) *All signals $X_1^{\sigma_1, \tau_1}, \dots, X_n^{\sigma_n, \tau_n}$ are mutually independent given S ;*
- (ii) *$f_S^{\sigma, \tau}(x|s')/f_S^{\sigma, \tau}(x|s)$ is increasing in x for every $s' > s$;*
- (iii) *$F^{\sigma, \tau}(x|s, t_i)$ is differentiable in σ, τ, x, s , and t_i .*

Assumption 4(i) requires signals to be independent conditional on the common component, (ii) imposes a strong MLRP in S , and (iii) allows a marginal approach.

Analysis.—Let $(\sigma, \tau, \beta_A^{\sigma, \tau})$ be a candidate equilibrium in auction format $A \in \{I, II\}$ in which bidders choose (σ, τ) and bid with a pure monotonic bidding function $\beta_A^{\sigma, \tau}(x)$, which constitutes a best-response bidding function to this information choice. In what follows, I assume that $\beta_A^{\sigma, \tau}(\cdot)$ is differentiable in x, σ , and τ . Without loss, consider bidder 1's incentives to deviate from this candidate equilibrium. Throughout this section, I am fixing the strategy $(\sigma, \tau, \beta_A^{\sigma, \tau})$ of all bidders but bidder 1, and therefore from now on suppressing their strategy in the notation.

Given a realization of the common component s , let $G(x|s)$ and $g(x|s)$ denote the distribution and density of the first-order statistic $\max_{j=2, \dots, N} \{X_j^{\sigma, \tau}\}$ of the $N - 1$ signals of all bidders but bidder 1 with information choice (σ, τ) . By assumption, this first-order statistic is independent of the private component T_1 of bidder 1. The expected payoff of bidder 1 when placing a bid b when $S = s$ and $T_1 = t_1$ does not depend on her information choice (σ_1, τ_1) or her realized signal x and can be expressed for the FPA and the SPA as

$$(1) \quad u_I(s, t_1, b) := [\psi(s, t_1) - b] G(\beta_I^{\sigma, \tau}(b)^{-1} | s),$$

$$(2) \quad u_{II}(s, t_1, b) := \int_{\beta_{II}^{\sigma, \tau}(b)^{-1}}^{\beta_{II}^{\sigma, \tau}(b)^{-1}} [\psi(s, t_1) - \beta_{II}^{\sigma, \tau}(y)] g(y|s) dy.$$

If bidder 1 follows the same symmetric strategy $(\sigma, \tau, \beta_A^{\sigma, \tau})$ as all other bidders, then she wins if and only if she has the highest signal of all bidders in either auction. In this case, her payoff difference between the FPA and the SPA takes a simple and

²¹ Note that Assumption 3 does *not* imply any ranking of $X_i^{\sigma', \tau}$ and $X_i^{\sigma, \tau}$ ($X_i^{\sigma', \tau}$ and $X_i^{\sigma, \tau}$) in terms of their accuracy about T_i (about S). See Examples 2 and 3 for different implications of higher accuracy about one component for the other component.

²² The same support assumption is for simplicity of notation; results can be easily extended to a framework where the support of an experiment depends on the information choice (σ, τ) .

tractable form because it no longer depends on her private component, as Lemma 1 shows.²³

LEMMA 1: For all s and x , $u_I(s, t_1, \beta_A^{\sigma, \tau}(x)) - u_{II}(s, t_1, \beta_A^{\sigma, \tau}(x))$ is constant in t_1 .

Bidder 1 wins the object worth $\psi(s, t_1)$ with the same winning probability $G(x|s)$ in the FPA and the SPA. For any signal realization x , the private-component realization t_1 affects neither the winning probability nor the expected payment because it is independent from the others' bids. Hence, in any symmetric monotonic candidate equilibrium, the difference in the expected payoff between the SPA and the FPA stems from the expected payment, which depends only on the realization of the common component, not the private one. Clearly, the private component is payoff relevant for both auction formats in isolation. However, in a symmetric environment, the private component does not enter the expected payment in which the two auction formats vary.

Bidder 1's expected payoff in auction $A \in \{I, II\}$ with information choice (σ_1, τ_1) is

$$R^A(\sigma_1, \tau_1) := \int_S \int_T \int_{\mathcal{X}} u_A(s, t_1, \beta_A^{\sigma_1, \tau_1}(x)) f^{\sigma_1, \tau_1}(x|s, t_1) dx dH_T(t_1) dH_S(s) - c(\sigma_1, \tau_1),$$

where $\beta_A^{\sigma_1, \tau_1}(x)$ is bidder 1's best-response bid with signal $X_1^{\sigma_1, \tau_1} = x$ when $N - 1$ other bidders play $(\sigma, \tau, \beta_A^{\sigma, \tau})$. In a candidate equilibrium $(\sigma, \tau, \beta_A^{\sigma, \tau})$ in auction format $A \in \{I, II\}$, bidder 1's **marginal revenue** of learning more about

- the private component is denoted by $MR_\tau^A(\sigma, \tau) := \left. \frac{\partial R^A(\sigma, \tau_1)}{\partial \tau_1} \right|_{\tau_1 = \tau}$,
- the common component is denoted by $MR_\sigma^A(\sigma, \tau) := \left. \frac{\partial R^A(\sigma_1, \tau)}{\partial \sigma_1} \right|_{\sigma_1 = \sigma}$.

Using this notation, I compare the marginal revenue from learning about either of the two components in the two auctions at the same information choice (σ, τ) .

Choosing the Common-Component Accuracy σ_1 .—The following result shows that bidders have higher incentives to learn about the common component in the FPA than in the SPA.

PROPOSITION 3: For any τ and σ , $MR_\sigma^I(\sigma, \tau) \geq MR_\sigma^{II}(\sigma, \tau)$.

If bidder 1 deviates from the candidate equilibrium by increasing σ_1 , this has two effects: (i) it increases the accuracy of her signal about the common component, and (ii) it might also vary the information about her private component in her signal. The example below depicts one instance where both these effects are present.

²³This does not hold for asymmetric equilibria where a bidder might win against different types depending on the auction format, or for different information choices in the SPA and the FPA.

EXAMPLE 2: Let all components be distributed with $S, T_1, \dots, T_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$. With information choice (σ_i, τ_i) , bidder i learns a signal $X_i^{\sigma_i \tau_i} = \sigma_i S + \tau_i T_i + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, 1)$. Then, an increase in σ_i corresponds to a more accurate signal about the common component and less accurate signal about the private one. The reverse is true for τ_i .

It is instructive to first discuss effect (i) in isolation: consider a simplified CV model where there is no payoff-relevant private component and bidders choose the accuracy of their common-component signal by choosing σ . For this one-dimensional learning model, Persico (2000) has shown that bidders have higher incentives to learn in the FPA than in the SPA.²⁴ This is because a winning bidder pays her own bid in the FPA as opposed to the second-highest bid in the SPA; so, having more information about other bidders (via a higher accuracy about the common component) is more valuable in the FPA than the SPA since this helps to bid closer to the second-highest bid and not to “leave money on the table.” This insight extends to my learning environment.

However, in my two-dimensional learning framework, there is also the additional effect (ii) present. Does a change in information about the private component from an increase in σ_1 affect the SPA and the FPA differently? It turns out that there is an unambiguous answer to this question in a symmetric candidate equilibrium: the effect on the private-component information of a marginal increase in σ_1 has identical payoff implications in the SPA and the FPA. For this insight, the intuition of Lemma 1 proves essential. Evaluated at a symmetric candidate equilibrium with (σ, τ) , the difference between the SPA and the FPA stems exclusively from the expected payment, not the expected object value. And even though a deviation in σ_1 affects the information about the private component in addition to that about the common one, it does not affect the expected payment. This insight allows me to extend the techniques from Persico (2000) to my two-dimensional learning model and establish Proposition 3 absent *any* assumption on how σ_1 affects the information about the private component.

Proposition 3 provides a marginal statement. As in Persico (2000), it is straightforward to extend it to an equilibrium statement when the accuracy of the private signal τ is given, and bidders only choose σ . Assume that there exists a unique symmetric monotonic equilibrium in the SPA and the FPA, and let σ^I and σ^{II} denote the equilibrium accuracy about the common component in the FPA and SPA, respectively. Moreover, assume that $MR_\sigma^I(\sigma, \tau)$ and $MR_\sigma^{II}(\sigma, \tau)$ cross zero once from above, at σ^I (for the FPA) and σ^{II} (for the SPA), respectively. Then, $\sigma^I \geq \sigma^{II}$.

Choosing the Private-Component Accuracy τ_1 .—Next, I determine whether increasing accuracy about the private component by increasing τ_1 is more profitable in the FPA or the SPA. An increase in τ_1 might affect the information content of the signal not only about the private component but also the common one. The previous insights conveyed the importance of accuracy about S for learning incentives.

²⁴Persico (2000) considers a general framework in which bidders choose their accuracy about a one-dimensional component and the bidders' components are affiliated. A crucial difference to my environment, however, is that bidders only learn about a one-dimensional component in his framework.

Hence, to pin down the marginal effect of increasing τ_1 , the following result imposes further assumptions on the impact of τ_1 on accuracy about S .²⁵

PROPOSITION 4: *For any accuracy about the common component σ ,*

1. *if $X_i^{\sigma, \tau} \succ_S X_i^{\sigma, \tau'}$ for any $\tau \leq \tau'$, then $MR_\tau^H(\sigma, \tau) \geq MR_{\tau'}^I(\sigma, \tau)$;*
2. *if $X_i^{\sigma, \tau} \prec_S X_i^{\sigma, \tau'}$ for any $\tau \leq \tau'$, then $MR_\tau^H(\sigma, \tau) \leq MR_{\tau'}^I(\sigma, \tau)$;*
3. *if $X_i^{\sigma, \tau} \sim_S X_i^{\sigma, \tau'}$ for any τ, τ' , then $MR_\tau^H(\sigma, \tau) = MR_{\tau'}^I(\sigma, \tau)$.*

This proposition mirrors the same intuition as Proposition 3: a marginal increase in common-component accuracy is more beneficial in the FPA than the SPA, while a marginal change in private-component accuracy affects the FPA and the SPA equally. While in Proposition 3, these preceding two effects were driven by a change in σ_1 , here, they are driven by a change in τ_1 and its accuracy implications about S .

Example 2 illustrates case 1 of Proposition 4 and captures an arguably realistic trade-off between learning about two different aspects from one information source: by increasing τ_1 and learning with a higher accuracy about the private component, a bidder decreases the accuracy of her signal about S since this information gets crowded out by the additional information about T_1 . In this case, learning more about the private component is more valuable in the SPA than the FPA.

Example 3 with $k = 1$ captures case 3 of Proposition 4. The marginal distribution $F_S^{\sigma, \tau_1}(\cdot | s)$ is identical for any τ_1 . So, the joint distribution of bidders' signals given the common component is unaffected by τ_1 . It has been shown in the literature that the incentives to acquire information coincide in the FPA and the SPA in an IPV environment (see, e.g., Hausch and Li 1991; Persico 2000). My results help view this insight in a wider perspective: independence of the private signals is not necessary for this equivalence in information acquisition in the FPA and the SPA. Rather, if learning does not affect the joint distribution of signals given S , learning incentives will coincide in the SPA and the FPA even if signals are correlated.²⁶

EXAMPLE 3: *Let all components be distributed with $S \sim \mathcal{N}(0, 1)$ and $T_1, \dots, T_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \Sigma)$, and fix some $k \geq 1$. With information choice (σ_i, τ_i) , bidder i observes the sum of two independent signals $X_i^{\sigma_i, \tau_i} = X_i^{S, \sigma_i} + X_i^{T_i, \tau_i}$, where $X_i^{S, \sigma_i} | S \sim \mathcal{N}(\sigma_i S, k(1 - \sigma_i^2))$ is a signal exclusively about S and $X_i^{T_i, \tau_i} | T_i \sim \mathcal{N}(\tau_i T_i, k\Sigma(1 - \tau_i^2))$ is a signal exclusively about T_i . For $k = 1$, the marginal distributions are $X_i^{S, \sigma_i} \sim \mathcal{N}(0, 1)$ and $X_i^{T_i, \tau_i} \sim \mathcal{N}(0, \Sigma)$ for any information choice (σ_i, τ_i) . Hence, varying σ_i or τ_i does not affect the normal noise that is added to the private- or common-component signal, respectively. For $k > 1$, increasing σ_i (τ_i) also*

²⁵ The cases of Proposition 4 are not exhaustive. One can find examples where increasing τ_1 changes the information about S in a manner that cannot be ranked in terms of $\succ_S$.

²⁶ A related insight for a one-dimensional learning environment can be found in Proposition 6 in Matthews (1977, p. IV-3): in a CV model, he finds a particular assumption on the conditional distribution function of the common-component signals such that information acquisition incentives in a symmetric candidate equilibrium are identical in the FPA and the SPA.

decreases the noise that is added to the private-component (common-component) signal, leading to a more accurate signal about both components.

Strikingly, in Example 3 with $k > 1$, bidders have higher incentives to learn about the private component by increasing τ_i in the FPA than the SPA, despite the fact that the private component is independent from the valuation and signals of all other bidders. This is because higher accuracy about T_i goes hand-in-hand with higher accuracy about S since it reduces mutual noise in the signal, and case 2 of Proposition 4 applies.

Similarly to the insights for choosing the common-component accuracy, it is straightforward to extend each of the three cases of Proposition 4 to an equilibrium prediction for a given σ under additional assumptions, e.g., about the marginal costs.

Two-Dimensional Information Choice.—Importantly, in the two-dimensional learning problem when bidders choose both σ and τ , how does the equilibrium information choice in the SPA and the FPA differ? Consider the arguably more realistic case where learning more about the private component comes at the opportunity costs of learning less about the common one, case 1 of Proposition 4.

COROLLARY 1: *Let $X_i^{\sigma,\tau} \succ_S X_i^{\sigma,\tau'}$ for any $\tau \leq \tau'$ and any σ . Let $c(\sigma, \tau)$ be such that for both auction formats $A \in \{I, II\}$, $MR_\sigma^A(\sigma, \tau)$ is continuous and nonincreasing in τ for all σ , $MR_\tau^A(\sigma, \tau)$ is continuous and nonincreasing in σ for all τ , and $[-MR_k^A(\sigma, \tau)]$ is strictly pseudomonotone²⁷ in $k \in \{\sigma, \tau\}$ for all σ and τ . Assume that there exists a unique symmetric equilibrium in the FPA and the SPA with information choice (σ^I, τ^I) and (σ^{II}, τ^{II}) , respectively. Then,*

$$\sigma^I \geq \sigma^{II} \quad \text{and} \quad \tau^I \leq \tau^{II}.$$

The assumptions above guarantee that for every τ , $MR_\sigma^A(\sigma, \tau)$ crosses zero at most once, from above, and at a point that is nonincreasing in τ . Similarly, for all σ , $MR_\tau^A(\sigma, \tau)$ crosses zero at most once, from above, and at a point that is nonincreasing in σ . This, together with Propositions 3 and 4, implies that at every τ , bidders choose a higher common-component accuracy in the FPA than in the SPA, and at every σ , bidders choose a higher private-component accuracy in the SPA than in the FPA. Combining this with equilibrium uniqueness yields the learning prediction above.

So, under the assumptions of Corollary 1, which auction format—FPA or SPA—leads bidders to acquire a more accurate signal about their private components, and to a more efficient allocation? Answering this requires additional assumptions since under the above conditions of Corollary 1, the accuracy relation about T_i among experiments is incomplete.²⁸

²⁷ I.e., $MR_\sigma^A(\sigma, \tau) \leq 0$ implies $MR_\sigma^A(\sigma', \tau) < 0$ for all $\sigma' > \sigma$ and for all τ , and $MR_\tau^A(\sigma, \tau) \leq 0$ implies $MR_\tau^A(\sigma, \tau') < 0$ for all $\tau' > \tau$ and for all σ .

²⁸ The assumptions of Corollary 1 do not specify whether $X_i^{\sigma,\tau}$ and $X_i^{\sigma',\tau}$ for $\sigma \neq \sigma'$ can be ranked in terms of their accuracy about T_i .

PROPOSITION 5: *Let the assumptions of Corollary 1 be satisfied. If for any $\sigma \leq \sigma'$ and any τ , $X_i^{\sigma,\tau} \succ_{T_i} X_i^{\sigma',\tau}$, then*

$$X_i^{\sigma',\tau'} \succ_S X_i^{\sigma'',\tau''} \quad \text{and} \quad X_i^{\sigma'',\tau''} \succ_{T_i} X_i^{\sigma',\tau'}.$$

Moreover, if for every realization of S , $X_i^{\sigma,\tau'} \succ_{T_i} X_i^{\sigma,\tau}$ and $X_i^{\sigma,\tau} \succ_{T_i} X_i^{\sigma',\tau}$ for any $\sigma' \geq \sigma$ and any $\tau' \geq \tau$, then for any realized vector of components $\{S, T_1, \dots, T_n\}$, the bidder with the highest private component $\max_i T_i$ is more likely to win in the SPA than in the FPA.

The first statement of Proposition 5 provides sufficient conditions for the SPA to lead to a higher accuracy about the private component and a lower accuracy about the common component than the FPA. This follows immediately from Corollary 1 and the transitivity of the accuracy relation (proven in Lemma 6): $X_i^{\sigma'',\tau''} \succ_{T_i} X_i^{\sigma',\tau'} \succ_{T_i} X_i^{\sigma,\tau}$ and $X_i^{\sigma,\tau} \succ_S X_i^{\sigma'',\tau''} \succ_S X_i^{\sigma',\tau'}$.

The second statement of Proposition 5 establishes a strong form of allocative efficiency: for any common-component realization, the bidder with the highest private component is more likely to receive the highest signal of all—and therefore win—in the SPA than in the FPA. A sufficient condition for this to be true is to strengthen the accuracy relation $\succ_{T_i}$ pointwise for every S .²⁹

The discussion above generalizes the efficiency insights for the SPA from the motivating example and Proposition 2 to a costly two-dimensional learning problem for the case wherein there is a trade-off between learning about the two components. In contrast, if there is no trade-off between learning either component, and learning about either component is a substitute for learning about the other, then the efficiency advantage of the SPA over the FPA can reverse. Strikingly, for case 2 of Proposition 4, the FPA might lead to a *higher* accuracy about the private component than the SPA. This is because an increase in τ goes hand-in-hand with a higher accuracy about the common component. Consequently, bidders might find it cheaper to learn about S by means of increasing τ rather than σ . The example below starkly demonstrates that the FPA can be strictly more efficient than the SPA. In the FPA, the bidder with the highest valuation wins with certainty (highest feasible allocative efficiency), while the SPA has the same meager efficiency properties as a random lottery: in the unique pure CV equilibrium, bidders prefer not to learn at all about their private components.

EXAMPLE 4: *Assume that there are two bidders with log-additive valuations³⁰ $V_i(S, T_i) = \exp(S + T_i)$. The components and signals are distributed as in Example 3 with $\Sigma = 1/3$ and $k = 3/2$. For simplicity, let $(\sigma_i, \tau_i) \in \{0, 1\}^2$ and $c(\sigma_i, \tau_i) = \tau_i/2$: bidders choose between perfectly revealing or uninformative signals to be added together, X_i^{S, σ_i} and X_i^{T, τ_i} . Since choosing $\sigma_i = 1$ is costless and*

²⁹This assumption is satisfied by the two previously discussed Examples 2 and 3.

³⁰This valuation specification is used, e.g., in Li, Perrigne, and Vuong (2000) for OCS wildcat auctions. See also Heumann (2019) for two-dimensional signals in ascending auctions.

thus, optimal, in any equilibrium bidders choose between the following two experiments $X_i^{\sigma_i, \tau_i}$:

$$\begin{aligned} X_i^{1,0} &= S + \epsilon_i, \quad \text{where } \epsilon_i \sim \mathcal{N}\left(0, \frac{1}{2}\right), \\ X_i^{1,1} &= S + T_i, \quad \text{where } T_i \sim \mathcal{N}\left(0, \frac{1}{3}\right). \end{aligned}$$

Crucially, learning more about the private component by choosing $\tau_i = 1$ over $\tau_i = 0$ also leads to more information about the common component (by decreasing the variance of the noise added to S from $1/2$ to $1/3$). This is more profitable in the FPA than in the SPA, as it conveys more information about the signal (and bid) of the other bidder.

The main observation (proven in the online Appendix) is that in the unique symmetric pure-strategy equilibria of the SPA and the FPA, we have

$$(\sigma^I, \tau^I) = (1, 1) \quad \text{and} \quad (\sigma^{II}, \tau^{II}) = (1, 0).$$

What does this observation imply for an efficiency comparison between the two auction formats? The SPA leads to a pure CV equilibrium where no bidders possess any information about their private component; starkly, the SPA has the same efficiency properties as a random lottery. The FPA, on the other hand, achieves the highest feasible allocative efficiency: the bidder with the highest value wins with probability one.

IV. Discussion

This paper explores the intricate link between the choice of an auction format and the endogenous interdependence in private information of bidders. Information about the common component simplifies coordination and is informative about the other bidders' bids. However, it is socially wasteful when it comes at the opportunity cost of learning less socially valuable information about their own private component. If the auctioneer switches the auction format from the SPA to the FPA, bidders might react by learning more about the common component and less about the private one. In the worst-case scenario, bidders might learn exclusively about the common component, and the FPA achieves the same allocative efficiency as a random lottery. Yet this result might flip if learning about the private component is tied to learning about the common component: then, the FPA might be strictly more efficient than the SPA.

In this paper, I focused on two standard sealed-bid auction formats, SPA and FPA. An interesting and open question is how my results extend to the English auction, another standard and popular auction format. Since the English auction is equivalent to the SPA for the case of two bidders, all my results up to Section IIIB for the SPA also hold for the English auction: absent learning costs, the learning incentives push toward less correlation (and thus, IPV) by means of learning more about the private component. The case of more than two bidders is more involved. Assume that even in the dynamic English format, bidders can acquire information only before the auction. Then, in a classic one-dimensional learning environment,

Hernando-Veciana (2009) notes that the proof techniques in Persico (2000) can be extended to show that bidders have lower incentives to learn in the English auction than in the SPA. This is due to the additional information—the signals of other bidders dropping out—being revealed during the English auction, which in turn reduces the incentives to acquire information initially. Extending this logic to my two-dimensional learning choice, if there is a learning trade-off between the components, then bidders should learn less about the common component (about which the others' dropout time is informative) and more about the private component (about which the others' dropout time contains no information) in the English auction as opposed to the SPA. Thus, under appropriate conditions, one would expect the English auction to be more efficient than the SPA. An even more interesting open question arises when allowing bidders to acquire information sequentially in the English auction: How does the price sequence at which other bidders drop out impact when and about which components the remaining bidders learn?

Aside from showing the significance of learning incentives when choosing an auction format, another policy implication of my work concerns the disclosure of public information about the common component. It can be socially valuable if the social planner discloses precise public information about the common component: when there is little uncertainty about the common component, the value of learning further about it is low, and bidders instead divert their learning attention to their private components. Hence, in addition to the choice of an auction format, the disclosure of public information seems to be a powerful tool for the social planner. Finding the efficient public disclosure policy is left for future research.

More broadly, agents choose their private information in a variety of other strategic environments, for example, in voting and public good problems. Understanding how a mechanism impacts the type of information that agents acquire is essential if agents have multidimensional valuations with varying efficiency implications. Many models in these environments have well-known results for a CV and IPV framework; importantly, policy recommendations vary based on these two frameworks. Yet if the endogeneity of private information in response to a mechanism is not taken into account, then policy recommendations might backfire and lead to a loss in efficiency: agents might be driven to learn about aspects that do not matter for an efficient outcome. The question as to for which mechanisms an IPV or interdependent value framework arises endogenously in other strategic problems provides an exciting new avenue for future research.

APPENDIX A

A. Order Statistics and Stochastic Dominance

The following lemma establishes a strong stochastic ordering of variables with the same marginal distribution but varying correlation.

Let X_1 , $\hat{X}_1$, and X_2 be three continuous random variables, distributed identically with marginal distribution $F(x)$ for $x \in \mathcal{X}$. $\hat{X}_1$ and X_2 are drawn independently. X_1 and X_2 are drawn independently conditional on some random variable S with positive density $h(s)$. X_1 and X_2 satisfy the strong MLRP with respect to S .

Let $Y_{(2)} := \min\{X_1, X_2\}$ be the second-order statistic of X_1 and X_2 . Denote the respective distribution by $G_{(2)}$. Let $Y_{(1)} := \max\{X_1, X_2\}$ be the first-order statistic of X_1 and X_2 with distribution $G_{(1)}$. Similarly, define $\hat{Y}_{(2)} := \min\{\hat{X}_1, X_2\}$ as the second-order statistic of $\hat{X}_1$ and X_2 , with distribution $\hat{G}_{(2)}$. Let $\hat{Y}_{(1)} := \max\{\hat{X}_1, X_2\}$ be the first-order statistic of $\hat{X}_1$ and X_2 with distribution $\hat{G}_{(1)}$.

LEMMA 2: For any x such that $F(x) \in (0, 1)$,

(i) $\hat{G}_{(1)}$ first-order stochastically dominates $G_{(1)}$, and $\hat{G}_{(1)}(x) < G_{(1)}(x)$,

(ii) $G_{(2)}$ first-order stochastically dominates $\hat{G}_{(2)}$, and $\hat{G}_{(2)}(x) > G_{(2)}(x)$.

PROOF:

Proof of Part 1: The distribution of the first-order statistic of the two independent random variables, $\hat{Y}_{(1)}$, is $\hat{G}_{(1)}(x) = F(x)^2$. The distribution of the first-order statistic of the two interdependent variables, $Y_{(1)}$, is $G_{(1)}(x) = \int_S F(x|s)^2 h(s) ds$. For all x with $F(x) \in (0, 1)$, using the strict Cauchy-Bunyakovsky-Schwarz inequality and the strong MLRP,³¹

$$\hat{G}_{(1)}(x) = F(x)^2 = \left[\int_S F(x|s) h(s) ds \right]^2 < \underbrace{\int_S h(s) ds}_{=1} \int_S F(x|s)^2 h(s) ds = G_{(1)}(x).$$

Proof of Part 2: The distribution of the second-order statistic of two i.i.d. variables is $\hat{G}_{(2)}(x) = 2F(x) - F(x)^2$. The weighted first- and second-order statistic distributions add up to the marginal distribution, i.e., $F(x) = \frac{1}{2}G_{(1)}(x) + \frac{1}{2}G_{(2)}(x)$. Thus,

$$(A1) \quad G_{(2)}(x) = 2F(x) - G_{(1)}(x) = 2F(x) - \int_S F(x|s)^2 h(s) ds.$$

The same argument as above (the strict Cauchy-Bunyakovsky-Schwarz inequality and the strict MLRP property) yields the result. ■

B. Omitted Proofs

PROOF OF PROPOSITION 1:

The marginal distribution of $X_i^{\sigma_i}$ when $S = 0$ is $f_S^{\sigma_i}(x|S = 0) := \frac{1}{2}f^{\sigma_i}(x|S = 0, T_i = 1) + \frac{1}{2}f^{\sigma_i}(x|S = 0, T_i = 0) = 1 + \sigma_i(1 - 2x)$. The corresponding distribution is $F_S^{\sigma_i}(x|S = 0) = x[1 + \sigma_i(1 - x)]$. When $S = 1$, signal $X_i^{\sigma_i}$ has a density $f_S^{\sigma_i}(x|S = 1) = 1 - \sigma_i(1 - 2x)$ and a distribution $F_S^{\sigma_i}(x|S = 1) = x[1 - \sigma_i(1 - x)]$.

³¹The Cauchy-Bunyakovsky-Schwarz inequality $\left[\int_a^b c(s)d(s)ds \right]^2 \leq \int_a^b c(s)^2 ds \cdot \int_a^b d(s)^2 ds$ is strict unless $c(s) = \alpha \cdot d(s)$ for some constant α (see Hardy, Littlewood, and Pólya 1934, Chapter VI). In the above argument, $c(s) = \sqrt{h(s)}$, and $d(s) = F(x|s)\sqrt{h(s)}$. Due to the strong MLRP, unless x such that $F(x) \in \{0, 1\}$, $F(x|s)$ is not constant in s .

First, consider the candidate equilibrium (σ, β_I^σ) in the FPA. Let bidder 1 deviate to $(\sigma_1, \beta_I^\sigma)$ with $\sigma_1 > \sigma$, i.e., applying the same bidding function to signal $X_1^{\sigma_1}$ as prescribed by the candidate equilibrium for X_1^σ . So, bidder 1 wins if and only if she has a higher signal realization than bidder 2 (ignoring ties is without loss).

Bidder 1's *winning probability* is $1/2$ in the candidate equilibrium, as well as in any deviation in the above deviation, using the above expressions for distributions and densities:

$$\Pr(X_1^{\sigma_1} \geq X_2^\sigma) = \frac{1}{2} \int_0^1 [F_S^\sigma(x|S=0)f_S^{\sigma_1}(x|S=0) + F_S^\sigma(x|S=1)f_S^{\sigma_1}(x|S=1)]dx = \frac{1}{2}.$$

Bidder 1's *expected value of the object when winning* is also constant for any σ_1 . If $V_1 = 0$, then $S = T_1 = 0$, and signal $X_1^{\sigma_1}$ has density $f^{\sigma_1}(x|S=0, T_1=0) = 2 - 2x$ for all σ_1 . Analogously, if $V_1 = 2$, then $S = T_1 = 1$, and again, σ_1 does not affect the signal's density. Finally, since overall winning probability is $1/2$ for any σ_1 and there is no change in winning probability for $V_1 \in \{0, 2\}$, bidder 1 must win with constant probability for the only remaining value realization $V_1 = 1$ for any σ_1 .

Next, I show that the *expected payment of bidder 1 conditional on winning* is decreasing in σ_1 if $\sigma > 0$. Using the above expressions for F_S^σ and f_S^σ , bidder 1's signal distribution conditional on having a higher signal than bidder 2 is

$$G_{(1)}^{\sigma_1, \sigma}(x) := \Pr(X_1^{\sigma_1} \leq x | X_1^{\sigma_1} \geq X_2^\sigma) = [1 + \sigma_1 \sigma (1 - x)^2]x^2.$$

Note that $G_{(1)}^{\sigma_1, \sigma}$ is increasing in σ_1 : for all $x \in (0, 1)$ and $\sigma > 0$, $G_{(1)}^{\sigma_1, \sigma}(x) > G_{(1)}^{\sigma, \sigma}(x)$ if $\sigma_1 > \sigma$. Hence, bidder 1's expected payment conditional on winning with the above deviation strategy in the FPA is smaller than in the candidate equilibrium since the bidding function is increasing: $\int_0^1 \beta_I^\sigma(x) dG_{(1)}^{\sigma_1, \sigma}(x) < \int_0^1 \beta_I^\sigma(x) dG_{(1)}^{\sigma, \sigma}(x)$.

In sum, $(\sigma_1, \beta_I^\sigma)$ for $\sigma_1 > \sigma > 0$ is a strictly profitable deviation since it strictly reduces the expected payment without affecting either the winning probability or the object value when winning. It was constructed using a tractable but suboptimal bidding function β_I^σ . Hence, $R^I(\sigma_1, \sigma)$ —the expected utility of bidder 1 after learning $\sigma_1 > \sigma > 0$ and bidding optimally—is weakly higher than the above deviation payoff, establishing $R^I(\sigma_1, \sigma) > R^I(\sigma, \sigma)$. Hence, no equilibrium with $\sigma \in (0, \bar{\sigma})$ can exist.

Second, consider the candidate equilibrium $(\sigma, \beta_{II}^\sigma)$ in the SPA. Let bidder 1 deviate to $(\sigma_1, \beta_{II}^\sigma)$ with $\sigma_1 < \sigma$. By the same argument as for the FPA, this deviating strategy has no effect on winning probability and on the value of the object in the event wherein bidder 1 wins. Let $G_{(2)}^{\sigma_1, \sigma}(x) := \Pr(X_2^\sigma \leq x | X_1^{\sigma_1} \geq X_2^\sigma)$ be bidder 2's signal distribution conditional on losing against bidder 1. A quick calculation reveals that $G_{(2)}^{\sigma_1, \sigma}$ is decreasing in σ_1 : for all $x \in (0, 1)$, $G_{(2)}^{\sigma_1, \sigma}(x) > G_{(2)}^{\sigma, \sigma}(x)$ if $\sigma_1 < \sigma$. Hence, bidder 1's expected payment (bidder 2's losing bid) conditional on winning with the above deviation strategy in the SPA is smaller than in the candidate equilibrium since the bidding function is increasing: $\int_0^1 \beta_{II}^\sigma(x) dG_{(2)}^{\sigma_1, \sigma}(x) < \int_0^1 \beta_{II}^\sigma(x) dG_{(2)}^{\sigma, \sigma}(x)$. In sum, $(\sigma_1, \beta_{II}^\sigma)$ with $\sigma_1 < \sigma$ constitutes a strictly profitable

deviation and yields a payoff weakly below the optimal deviation payoff: $R''(\sigma_1, \sigma) > R''(\sigma, \sigma)$. Thus, there exists no equilibrium in the SPA with $\sigma > \underline{\sigma}$. ■

PROOF OF PROPOSITION 2:

Assume by contradiction that there exists a candidate equilibrium where bidders learn $X'_i \neq X_i^T$ and follow an increasing bidding function $\beta(X'_i)$. I show that bidder 1 has a strictly profitable deviation: learn X_1^T and bid $\hat{\beta}(X_1^T) = \beta(X'_1)$. Resembling the Proof of Proposition 1, I show that in comparison to the candidate equilibrium, this deviation yields (i) the same winning probability, (ii) a strictly lower expected payment, and (iii) a weakly higher expected object value when winning.

In the symmetric candidate equilibrium, winning probability is $1/2$. In the deviation strategy, bidder 1 wins if $X_1^T \geq X'_2$. By Assumption 2, these two random variables are i.i.d. Hence, winning probability with the deviation is $1/2$.

In the candidate equilibrium, both bidders have the same marginal bid distribution, which is correlated via the common component. Hence, the expected payment of bidder 1 when learning X'_1 and winning is $\int_0^1 \beta(x) dG_{(2)}(x)$, where $G_{(2)}(x)$ is the second-order statistic distribution of two correlated identically distributed signals X'_1 and X'_2 as derived in (A1). Following bidder 1's deviation, bidders have the same marginal bid distributions as in the candidate equilibrium, but the bids are now independent. Hence, when learning X_1^T and winning, bidder 1 pays $\int_0^1 \beta(x) d\hat{G}_{(2)}(x)$, where $\hat{G}_{(2)}$ is the distribution of the second-order statistic of two i.i.d. variables (with the same marginal distribution as those in $G_{(2)}(x)$). The distribution of the second-order statistic under correlation strictly first-order stochastically dominates that under independence for the same marginal distribution. This is a general statistical property, which I establish in Lemma 2. A winning bidder pays the second-order statistic of the two bids. Hence, expected payment conditional on winning is strictly lower in the deviation.

Let bidder 1's net difference in expected private component when winning in the deviation (D) versus candidate equilibrium (C) be $\Delta_1^T := E[T_1 | 1 \text{ wins}, D] - E[T_1 | 1 \text{ wins}, C]$. Analogously, for the common component, let $\Delta_i^S := E[S | i \text{ wins}, D] - E[S | i \text{ wins}, C]$. The object is always sold, and both bidders win with probability $1/2$. Therefore,

$$\begin{aligned} E[S] &= \frac{1}{2}E[S | 1 \text{ wins}, C] + \frac{1}{2}E[S | 2 \text{ wins}, C] = \frac{1}{2}E[S | 1 \text{ wins}, D] \\ &\quad + \frac{1}{2}E[S | 2 \text{ wins}, D]. \end{aligned}$$

Hence, $\Delta_1^S = -\Delta_2^S$. Using this observation, I prove in the following that bidder 1's deviation is strictly profitable because $\Delta_1^T + \Delta_1^S = \Delta_1^T - \Delta_2^S \geq 0$.

In the candidate equilibrium, due to symmetry, both bidders win at every s with probability $1/2$. Hence, $E[S | i \text{ wins}, C] = E[S]$, and

$$(A2) \quad \Delta_1^T - \Delta_2^S = E[T_1 | 1 \text{ wins}, D] - E[T_1 | 1 \text{ wins}, C] - E[S | 2 \text{ wins}, D] + E[S].$$

Let $g_1(x)$ be the density of the first-order statistic of the two candidate equilibrium signals X'_1 and X'_2 . Conditional on winning in the candidate equilibrium, bidder 1's signal when winning is distributed with density g_1 . We can write

$$(A3) \quad \begin{aligned} E[T_1 | 1 \text{ wins}, C] &= \int E[T_1 | X'_1 = x] g_1(x) dx \\ &\leq \int E[T_1 | X'_1 = x] 2f(x) F(x) dx, \end{aligned}$$

where $2f(x)F(x)$ is the density of the first-order statistic of two i.i.d. random variables with marginal distribution $F(x)$. The inequality in the last step followed from Lemma 2: the first-order statistic of two i.i.d. variables first-order stochastically dominates that of correlated variables with the same marginal distribution, and $E[T_1 | X'_1 = x]$ is nondecreasing in x . Because $F(x)$ is the marginal distribution of any signal in $\mathcal{E}$, the next three expressions follow:

$$\begin{aligned} E[T_1 | 1 \text{ wins}, D] &= \int E[T_1 | X_1^T = x] 2f(x) F(x) dx; \\ E[S | 2 \text{ wins}, D] &= \int E[S | X'_2 = x] 2f(x) F(x) dx; \\ E[S | 2 \text{ wins}, C] &= E[S] = \int E[S] 2f(x) F(x) dx. \end{aligned}$$

Plugging this and (A3) back into (A2) yields

$$\begin{aligned} \Delta_1^T - \Delta_2^S &\geq \int (E[T_1 | X_1^T = x] - E[T_1 | X'_1 = x] \\ &\quad - E[S | X'_2 = x] + E[S]) 2f(x) F(x) dx = 0. \end{aligned}$$

The last equality followed by symmetry, $E[V_1 | X'_1 = x] = E[V_2 | X'_2 = x]$, and because by the premise, $E[V_1 | X'_1 = x] = E[V_1 | X_1^T = x]$. ■

LEMMA 3: $E[T_1 | X_1^T \geq X_2^T] \geq E[T_1 | X'_1 \geq X'_2]$ for all $X'_i \in \mathcal{E}$.

PROOF:

By Assumptions 1 and 2, for all $x \in [0, 1]$,

$$E[T_1 | X_1^T = x] + E[S] = E[T_1 | X'_1 = x] + E[S | X'_1 = x].$$

Integrating both sides yields

$$\begin{aligned} &\int E[T_1 | X_1^T = x] 2f(x) F(x) dx + E[S] \\ &= \int E[T_1 | X'_1 = x] 2f(x) F(x) dx + \int E[S | X'_1 = x] 2f(x) F(x) dx. \end{aligned}$$

Since $F(x) \geq F(x)^2$ for all x , and since $E[S | X'_1 = x]$ is nondecreasing in x by the MLRP, it holds that $E[S] = \int E[S | X'_1 = x] f(x) dx \leq \int E[S | X'_1 = x] 2f(x) F(x) dx$. Therefore, since the distribution of bidder 1's winning signal when both learn X_1^T

first-order stochastically dominates the distribution of any other signal (Lemma 2), it holds that

$$\begin{aligned}
 E[T_1 | X'_1 \geq X'_2] &= \int E[T_1 | X'_1 = x] g_{(1)}(x) dx \\
 &\leq \int E[T_1 | X'_1 = x] 2f(x) F(x) dx \\
 &\leq \int E[T_1 | X_1^T = x] 2f(x) F(x) dx \\
 &= E[T_1 | X_1^T \geq X_2^T]. \blacksquare
 \end{aligned}$$

PROOF OF LEMMA 1:

Plugging in a bid $b = \beta_A^{\sigma, \tau}(x)$ for $A \in \{I, II\}$ into (1) and (2) and taking the difference immediately yields the result. ■

PROOF OF PROPOSITION 3:

Fix the information choice (σ, τ) of all other bidders but bidder 1 and assume that they bid $\beta_A^{\sigma, \tau}(x)$, which is the symmetric equilibrium bidding function for N bidders with information choice (σ, τ) . Denote the expected utility of bidder 1 with a signal $X_1^{\sigma_1, \tau_1} = x$, a bid b when the common component is s as

$$\tilde{u}_A^{\sigma_1, \tau_1}(s, b, x) := \int_{\mathcal{T}} u_A(s, t_1, b) h_T^{\sigma_1, \tau_1}(t_1 | x, s) dt_1.$$

In contrast to the one-dimensional learning model in Persico (2000), in my model this expected payoff depends not only on the common component s and the bid b but also on the signal realization x and the information choice (σ, τ) . This is because x and (σ, τ) both contain payoff-relevant information about the private component T_i .³²

Next, I show that bidder 1's marginal revenue from learning more about the common component is higher in the FPA than the SPA. Using the above notation, we can express the difference in marginal revenues as

$$\begin{aligned}
 MR_\sigma^I(\sigma, \tau) - MR_\sigma^{II}(\sigma, \tau) \\
 = \frac{d}{d\sigma_1} \int_S \int_{\mathcal{X}} [\tilde{u}_I^{\sigma_1, \tau}(s, \beta_I^{\sigma_1, \tau}(x), x) - \tilde{u}_{II}^{\sigma_1, \tau}(s, \beta_{II}^{\sigma_1, \tau}(x), x)] f^{\sigma_1, \tau}(x | s) dx h_S(s) ds \Big|_{\sigma_1 = \sigma}.
 \end{aligned}$$

We wish to show that this difference is nonnegative. For simplicity of notation, let $\Delta^{\sigma_1, \tau}(s, x) := \tilde{u}_I^{\sigma_1, \tau}(s, \beta_I^{\sigma_1, \tau}(x), x) - \tilde{u}_{II}^{\sigma_1, \tau}(s, \beta_{II}^{\sigma_1, \tau}(x), x)$. Because $X_1^{\sigma_1, \tau} \succ_S X_1^{\sigma, \tau}$, by Definition 1, there is a transformation $M_{\sigma, \sigma_1, s}(\cdot)$ so that $M_{\sigma, \sigma_1, s}(X_1^{\sigma, \tau})$ and $X_1^{\sigma_1, \tau} | s$ are distributed identically.³³ We can apply a change of variables and express the

³²In Persico (2000), the expected payoff in auction A is pinned down by the bid b and the only component about which bidder i 's signal is informative: a one-dimensional component v_i .

³³Because τ is constant throughout this proof, I suppress it in the notation of $M_{\sigma, \sigma_1, s}(\cdot)$. For brevity, the dependence of this transformation on the common component is captured in the subscript.

difference in marginal revenues as

$$\begin{aligned}
 (A4) \quad MR_{\sigma}^I(\sigma, \tau) - MR_{\sigma}^{II}(\sigma, \tau) &= \frac{d}{d\sigma_1} \int_S \int_X \Delta^{\sigma_1, \tau}(s, x) f^{\sigma_1, \tau}(x|s) dx h_S(s) ds \Big|_{\sigma_1=\sigma} \\
 &= \frac{d}{d\sigma_1} \int_S \int_X \Delta^{\sigma_1, \tau}(s, M_{\sigma, \sigma_1, s}(x)) f^{\sigma, \tau}(x|s) dx h_S(s) ds \Big|_{\sigma_1=\sigma} \\
 &= \int_S \int_X \frac{d}{d\sigma_1} \Delta^{\sigma_1, \tau}(s, M_{\sigma, \sigma_1, s}(x)) h_S^{\sigma, \tau}(s|x) ds f^{\sigma, \tau}(x) dx \Big|_{\sigma_1=\sigma}.
 \end{aligned}$$

Note that

$$(A5) \quad \frac{d}{d\sigma_1} \tilde{u}_A^{\sigma_1, \tau}(s, \beta_A^{\sigma_1, \tau}(M_{\sigma, \sigma_1, s}(x)), M_{\sigma, \sigma_1, s}(x)) \Big|_{\sigma_1=\sigma}$$

$$(A6) \quad = \frac{\partial \tilde{u}_A^{\sigma, \tau}(s, \beta_A^{\sigma, \tau}(x), x)}{\partial b} \left[\beta_A^{\sigma, \tau}(x) \frac{\partial M_{\sigma, \sigma_1, s}(x)}{\partial \sigma_1} \Big|_{\sigma_1=\sigma} + \frac{\partial \beta_A^{\sigma_1, \tau}(x)}{\partial \sigma_1} \Big|_{\sigma_1=\sigma} \right]$$

$$(A7) \quad + \frac{\partial \tilde{u}_A^{\sigma_1, \tau}(s, \beta_A^{\sigma_1, \tau}(x), x)}{\partial \sigma_1} \Big|_{\sigma_1=\sigma} + \frac{\partial \tilde{u}_A^{\sigma, \tau}(s, \beta_A^{\sigma, \tau}(x), x)}{\partial x} \frac{\partial M_{\sigma, \sigma_1, s}(x)}{\partial \sigma_1} \Big|_{\sigma_1=\sigma}.$$

Because $\beta_A^{\sigma, \tau}$ is the optimal symmetric bidding function for information choice (σ, τ) , the first-order conditions yield

$$(A8) \quad \int_S \frac{\partial \tilde{u}_A^{\sigma, \tau}(s, \beta_A^{\sigma, \tau}(x), x)}{\partial b} h_S^{\sigma, \tau}(s|x) ds = 0.$$

Furthermore, $\nu^{\sigma, \tau}(s, x) := u_I(s, t_1, \beta_I^{\sigma, \tau}(x)) - u_{II}(s, t_1, \beta_{II}^{\sigma, \tau}(x))$ is not a function of t_1 by Lemma 1, and hence,

$$\begin{aligned}
 (A9) \quad & \frac{\partial [\tilde{u}_I^{\sigma_1, \tau}(s, \beta_I^{\sigma_1, \tau}(x), x) - \tilde{u}_{II}^{\sigma_1, \tau}(s, \beta_{II}^{\sigma_1, \tau}(x), x)]}{\partial \sigma_1} \\
 &= \frac{\partial}{\partial \sigma_1} \int_{\mathcal{T}} \nu^{\sigma, \tau}(s, x) f^{\sigma_1, \tau}(t_1|x, s) dt_1 = 0.
 \end{aligned}$$

Finally, note that

$$(A10) \quad \frac{\partial \tilde{u}_A^{\sigma, \tau}(s, \beta_A^{\sigma, \tau}(x), x)}{\partial b} \beta_A^{\sigma, \tau}(x) + \frac{\partial \tilde{u}_A^{\sigma, \tau}(s, \beta_A^{\sigma, \tau}(x), x)}{\partial x} = \frac{d \tilde{u}_A^{\sigma, \tau}(s, \beta^{\sigma, \tau}(x), x)}{dx}.$$

Plugging (A5), (A8), (A9), and (A10) into (A4), we obtain

$$\begin{aligned}
 (A11) \quad MR_{\sigma}^I(\sigma, \tau) - MR_{\sigma}^{II}(\sigma, \tau) &= \int_X \int_S \frac{d \Delta^{\sigma, \tau}(s, x)}{dx} \frac{\partial M_{\sigma, \sigma_1, s}(x)}{\partial \sigma_1} \Big|_{\sigma_1=\sigma} h_S^{\sigma, \tau}(s|x) ds f^{\sigma, \tau}(x) dx.
 \end{aligned}$$

In what follows, I apply Lemma 1 from the Appendix of Persico (2000), the proof of which is omitted. For completeness, I include his formal statement below.

LEMMA 4 (Persico 2000, p. 147): *Let $J(\cdot)$ be a nondecreasing function and $H(\cdot)$ be a quasimonotone function, i.e., $s' > s$ and $H(s) > 0$ imply $H(s') \geq 0$. If for some measure μ on $\mathbb{R}$ it holds that $\int_{\mathcal{S}} H(s) d\mu(s) = 0$, then $\int_{\mathcal{S}} H(s) J(s) d\mu(s) \geq 0$.*

The application of the above lemma pointwise for every x in (A11) yields the required result, where $J(s) = \frac{\partial M_{\sigma, \sigma_1, s}(x)}{\partial \sigma_1} \Big|_{\sigma_1 = \sigma}$ and $H(s) = d\Delta^{\sigma, \tau}(s, x)/dx$. The necessary properties to apply this lemma are established below.

LEMMA 5: *The function $d\Delta^{\sigma, \tau}(s, x)/dx$ satisfies the following:*

- (1)
$$\int_{\mathcal{S}} \frac{d\Delta^{\sigma, \tau}(s, x)}{dx} h_{\mathcal{S}^{\sigma, \tau}}^{\sigma, \tau}(s|x) ds = 0.$$
- (2) $\partial \Delta^{\sigma, \tau}(s, x)/\partial x$ is quasimonotone in s : $s' > s$ and $\partial \Delta^{\sigma, \tau}(s, x)/\partial x > 0$ imply $\partial \Delta^{\sigma, \tau}(s', x)/\partial x \geq 0$.

PROOF:

Proof of 1: The first-order conditions in (A8) and the notation $\nu^{\sigma, \tau}(s, x)$ yield

$$\begin{aligned} & \int_{\mathcal{S}} \frac{d\Delta^{\sigma, \tau}(s, x)}{dx} h_{\mathcal{S}^{\sigma, \tau}}^{\sigma, \tau}(s|x) ds \\ &= \int_{\mathcal{S}} \frac{\partial [\tilde{u}_I^{\sigma, \tau}(s, \beta^{\sigma, \tau}(x), x) - \tilde{u}_{II}^{\sigma, \tau}(s, \beta^{\sigma, \tau}(x), x)]}{\partial x} h_{\mathcal{S}^{\sigma, \tau}}^{\sigma, \tau}(s|x) ds \\ &= \int_{\mathcal{S}} \nu^{\sigma, \tau}(s, x) \int_{\mathcal{T}} \frac{\partial h_{\mathcal{T}^{\sigma, \tau}}^{\sigma, \tau}(t|x, s)}{\partial x} dt h_{\mathcal{S}^{\sigma, \tau}}^{\sigma, \tau}(s|x) ds = 0, \end{aligned}$$

where the last step followed because $h_{\mathcal{T}^{\sigma, \tau}}^{\sigma, \tau}$ is a density.

Proof of 2: Using the expressions for expected payoffs in equations (1) and (2), we obtain

$$\Delta^{\sigma, \tau}(s, x) = \nu^{\sigma, \tau}(s, x) = \int_{-\infty}^x [\beta_{II}^{\sigma, \tau}(y) - \beta_I^{\sigma, \tau}(x)] g(y|x) dy.$$

Due to Lemma 1, the expected object value drops out in the expression, the private component plays no role in this difference, and the proof from here proceeds as in the one-dimensional learning framework in Persico (2000):

$$\frac{\partial \Delta^{\sigma, \tau}(s, x)}{\partial x} = \left[\beta_{II}^{\sigma, \tau}(x) - \beta_I^{\sigma, \tau}(x) - \frac{\partial \beta_I^{\sigma, \tau}(x)}{\partial x} \frac{G(x|x)}{g(x|x)} \right] g(x|x).$$

By Assumption 4, all bidders' signals satisfy the MLRP with respect to S for any information choice (σ, τ) . Thus, $G(x|x)/g(x|x)$ is nonincreasing in s . Then, as s increases, this expression crosses zero at most once, from below. ■

As Persico (2000, p. 141) showed, $\left. \frac{\partial M_{\sigma, \sigma_1, s}(x)}{\partial \sigma_1} \right|_{\sigma_1 = \sigma}$ is nondecreasing in s . This is because $\left. \frac{\partial M_{\sigma, \sigma_1, s}(x)}{\partial \sigma_1} \right|_{\sigma_1 = \sigma} = \lim_{\sigma_1 \downarrow \sigma} \frac{M_{\sigma, \sigma_1, s}(x) - M_{\sigma, \sigma, s}(x)}{(\sigma_1 - \sigma)} = \lim_{\sigma_1 \downarrow \sigma} \frac{M_{\sigma, \sigma_1, s}(x) - x}{(\sigma_1 - \sigma)}$, $\sigma_1 \geq \sigma$, and $M_{\sigma, \sigma_1, s}(x)$ is nondecreasing in s . This observation and Lemma 5 allow the application of Lemma 4, and hence, $MR_{\sigma}^I(\sigma, \tau) - MR_{\sigma}^{II}(\sigma, \tau) \geq 0$. ■

REMARK 1 (Comparison to Persico 2000): *In Persico (2000), the payoff in auction A is pinned down by the bid b and the one-dimensional component s about which the bidder receives a signal. Beyond b and s , the accuracy of the signal and the signal realization are no longer payoff relevant. In contrast, in my framework the signal realization contains information about two components, S and T_i , which leads to a substantial complication over Persico's one-dimensional framework. The information choice (σ, τ) and the signal realization x remain payoff relevant even when fixing s and a bid b . This is because they contain payoff-relevant information about the private component T_i . Therefore, in contrast to Persico (2000), in my framework the expected payoff with s and bid b , denoted by $\tilde{u}_A^{\sigma, \tau}(s, b, x)$, depends on both x and σ, τ .*

As a consequence, there are two effects when learning more about the common component: (i) the effect of learning more about the common component and (ii) the effect that this has on the information content about the second value component—the private component. The former effect coincides with the forces laid out in Persico (2000), while the latter effect is absent in the one-dimensional learning framework in his paper. This second effect manifests in the additional line (A7) in the derivative with respect to σ_1 . However, it disappears as shown in (A9) because the effect of increasing σ_1 on the information content of T_1 has the same impact in the SPA and the FPA in a candidate equilibrium where a bidder wins if and only if she has a higher signal.

PROOF OF PROPOSITION 4:

By Proposition 3, higher accuracy about the common component is more valuable in the FPA than in the SPA. This holds irrespective of its implications for the information content about the private component because this change in information affects the FPA and the SPA equally. Hence, Proposition 3 yields the required result if the accuracy about S is parameterized by τ_1 .

By the premise of case 2 of the proposition, for $\tau_1 \geq \tau$, there exists a transformation $M_{\tau, \tau_1, s}$ such that $M_{\tau, \tau_1, s}(X_1^{\sigma, \tau})$ and $X_1^{\sigma, \tau_1} | s$ are distributed identically. Then, the proof proceeds exactly as in Proposition 3 after taking the derivative with respect to τ_1 instead of σ_1 , while fixing σ instead of τ . Case 1 is an immediate corollary of this. Finally, case 3 follows by combining cases 1 and 2. ■

PROOF OF PROPOSITION 5:

The first statement of Proposition 5 follows immediately from Corollary 1 and is explained in the main text. For the proof of the second statement, let $X_i^{\sigma', \tau'}$ and $X_i^{\sigma'', \tau''}$ be the equilibrium signals in the FPA and SPA.

Fix an arbitrary realized vector of all components, $\{s, t_1, \dots, t_N\}$. Without loss, assume that bidder 1 has the highest private component $t_1 \in \max\{t_1, \dots, t_N\}$. Let

$F^I(x|t_i, s)$ and $F^{II}(x|t_i, s)$ be the distribution of each bidder's signal in the FPA and SPA, respectively, when $S = s$ and $T_i = t_i$. By Corollary 1, $\tau^I \leq \tau^{II}$ and $\sigma^I \geq \sigma^{II}$, and by assumption, $X_i^{\sigma^{II}, \tau^{II}} \succ_{T_i} X_i^{\sigma^I, \tau^{II}} \succ_{T_i} X_i^{\sigma^I, \tau^I}$ for every s . So, using Definition 1, the function $M_s(x|t_i)$ such that $F^{II}(M_s(x|t_i)|s, t_i) = F^I(x|s, t_i)$ is nondecreasing in t_i . That is, $M_s(x|t_i)$ is the transformation mapping the less precise signal into a more precise signal with the same distribution given $T_i = t$ and $S = s$.

Next, using this fact that x and $M_s(x|t_i)$ are distributed identically given (s, t_i) , I prove that percentile-by-percentile of the two signal distributions $X_i^{\sigma^{II}, \tau^{II}}$ and $X_i^{\sigma^I, \tau^I}$, bidder 1 is more likely to win in the SPA than the FPA. Bidder 1's winning probability in the FPA with signal realization x is $\prod_{i \neq 1} F^I(x|t_i, s)$. Bidder 1's winning probability in the SPA with the corresponding signal realization $M_s(x|t_1)$ is $\prod_{i \neq 1} F^{II}(M_s(x|t_1)|s, t_i)$. Since $M_s(x|t_i)$ is nondecreasing in t_i and $t_1 \geq t_i$, winning probability in the SPA is higher than in the FPA:

$$\prod_{i \neq 1} F^{II}(M_s(x|t_1)|t_i, s) \geq \prod_{i \neq 1} F^{II}(M_s(x|t_i)|t_i, s) = \prod_{i \neq 1} F^I(x|t_i, s). \blacksquare$$

LEMMA 6: If $X^a \succ_Z X^b$ and $X^b \succ_Z X^c$, then $X^a \succ_Z X^c$.

PROOF:

For any z , let $F_a(\cdot|z)$, $F_b(\cdot|z)$, and $F_c(\cdot|z)$ be the distributions of X^a , X^b , and X^c , respectively. By assumption, for all $y \in X^b$ and $x \in X^c$, the functions $M^a(y|z) := F_a^{-1}(F_b(y|z)|z)$ and $M^b(x|z) := F_b^{-1}(F_c(x|z)|z)$ are nondecreasing in z . To prove the lemma, observe that for any $z' \geq z$,

$$\begin{aligned} F_a^{-1}(F_c(x|z')|z') &= F_a^{-1}(F_b(M^b(x|z')|z')|z') \\ &\geq F_a^{-1}(F_b(M^b(x|z')|z)|z') \\ &\geq F_a^{-1}(F_b(M^b(x|z)|z)|z') \\ &= F_a^{-1}(F_c(x|z)|z), \end{aligned}$$

where the first inequality follows from $X^a \succ_Z X^b$ and the second inequality follows from $X^b \succ_Z X^c$ and by $F_b(\cdot|z)$ and $F_a^{-1}(\cdot|z)$ being nondecreasing. $\blacksquare$

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